

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: Multiple Dose Study of MK-2828 in Participants With Type 2 Diabetes (MK-2828-003) **Official Regulatory Title:** Multiple Dose Study of MK-2828 in Participants With Type 2 Diabetes **Short Title/Acronym:** MK-2828-003 **Protocol Number:** MK-2828-003 **NCT Number:** NCT07089784 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** MK-2828 (Oral Capsule) **Phase of Development:** Phase 1 **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the safety and tolerability of multiple doses of MK-2828 when administered to participants with Type 2 Diabetes (T2D). * **Secondary Objectives:** To evaluate the pharmacokinetics (PK) of MK-2828 in the body over time (measuring C₂₄ in plasma after Day 1 and Day 27), and to determine its effect on the amount of high-sensitivity C-reactive protein (hsCRP) in the blood compared to placebo.

2.2 Background & Rationale To investigate the safety, tolerability, and pharmacokinetic profile of MK-2828, a novel nucleotide-binding, leucine-rich, protein 3 inhibitor (NLRP3i) based therapy in patients with Type 2 Diabetes. Furthermore, measuring hsCRP (a protein linked to inflammation) will help understand if reducing inflammation via MK-2828 can aid in the management and treatment of T2D.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Placebo-controlled * **Blinding:** Double-blind (Placebo-matching) * **Randomization:** Randomized

3.2 Planned Enrollment & Duration * **Target N:** 64 participants * **Number of Sites:** Multiple sites in the United States (e.g., CA, FL, TX, MO) * **Estimated Study Start:** 22-Sep-2025 * **Estimated Primary Completion:** 25-Mar-2026 * **Total Duration:** ~8 weeks (Screening: 2-4 wks, Treatment: 4 wks, Follow-up: 2 wks)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. With the exception of type 2 diabetes mellitus (T2DM), is in generally good health. 2. Has a history of T2DM for ≥ 1 year at the time of screening based on self-reporting, managed with lifestyle modification alone or with stable doses of ≤ 3 oral anti-diabetic medications for at least 3 months. 3. Has a baseline HbA1C level between 6.5% and $\leq 10\%$ at the time of screening, and a BMI between 25 and 40 kg/m², inclusive, with a stable weight for at least 3 months.

4.2 Exclusion Criteria 1. History of significant multiple and/or severe allergies, anaphylaxis, or systemic hypersensitivity to MK-2828, other NLRP3i therapies, or placebo. 2. Has type 1 diabetes mellitus, secondary types of diabetes, or experienced complications of diabetes (e.g., ketoacidosis, unstable diabetic retinopathy, maculopathy). 3. Has a history of congestive heart failure (NYHA Class 3 or 4), myocardial infarction, uncontrolled arrhythmias, cardiac revascularization, unstable peripheral arterial disease, and/or stroke within 6 months of screening.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Management Pathway:** Daily oral administration of MK-2828 or placebo for 28 consecutive days. * **Dosage/Frequency:** MK-2828 or matching placebo taken once per day. * **Route of Administration:** Oral capsule. * **Duration of Treatment:** 28 days of daily treatment.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Up to 3 oral anti-diabetic medications (must remain on stable doses without adjustment during the study). * **Prohibited:** Previous or planned obesity treatment with surgery or a weight loss device.

6. Study Timeline & Visit Schedule

- **Onsite Visit Cadence:** Screening, Day 1, Day 27, and post-treatment follow-up.
- **Remote Monitoring Frequency:** Daily virtual visits during the 28-day treatment phase.
- **Checklist Review Interval:** Baseline, Day 1, Day 27/28, and End of Study.

Visit ID	Window (Days)	Key Activities/Procedures
1	2-4 weeks prior	Informed Consent, Medical History, Baseline HbA1C & Labs
2	Day 0	Randomization, First Dose
3	Days 1 to 26	Daily visits (in-person or virtual), Safety AE Monitoring, PK Blood Collection (after Day 1 dose)
4	Day 27/28	Final Dose, PK Blood Collection (C24 after Day 27 dose), hsCRP Labs, IP Accountability, Follow-up (2 weeks post-treatment)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Incidence of Adverse Events (AEs) to assess the safety and tolerability of MK-2828 over the 28-day treatment period. * **Measurement Method:** Clinical observation, participant reporting, and safety laboratory assessments.

7.2 Secondary & Exploratory Endpoints * **Pharmacokinetics:** C24 of MK-2828 in plasma after the Day 1 and Day 27 doses. * **Biomarker:** Change from baseline in placebo-corrected high-sensitivity C-reactive protein (hsCRP).

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous monitoring of any untoward medical occurrence temporally associated with the use of study intervention (whether related or not) through in-person/virtual daily check-ins.
 - **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness.
 - **Data Monitoring Committee (DMC):** Internal sponsor committee managed by Merck Sharp & Dohme LLC.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Safety Analysis Set (all randomized participants who receive at least one dose) and Pharmacokinetic (PK) Analysis Set.
 - **Sample Size Justification:** $N = 64$ participants is powered appropriately for a Phase 1 safety, tolerability, and pharmacokinetic profiling trial.
 - **Handling of Missing Data:** Standard imputation or treated as missing completely at random for PK evaluation.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
- **Monitoring Plan:** Routine clinical site monitoring combined with daily remote/virtual check-ins with participants.
- **Quality Audit Mechanism:** Daily symptom tracking & study drug accountability monitoring.
- **Education Protocol (HCP):** Investigator training on Phase 1 safety monitoring and PK sampling windows.
- **Education Protocol (Patient):** Instructions for daily symptom logging and virtual visit adherence.

- **Audit Trail:** Standard tracking of eCRF entries and data clarification forms via Sponsor oversight.
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11. Study Identifiers & Governance

- **Primary ID:** {{MK-2828-003}}
 - **Registry Number:** {{NCT07089784}}
 - **Sponsor/Lead Organization:** {{Merck Sharp & Dohme LLC}}
 - **Study Title:** {{Multiple Dose Study of MK-2828 in Participants With Type 2 Diabetes (MK-2828-003)}}
 - **Official Regulatory Title:** {{Multiple Dose Study of MK-2828 in Participants With Type 2 Diabetes}}
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12. Clinical Framework (Type 2 Diabetes Specific)

- **Primary Condition:** {{Type 2 Diabetes Mellitus (T2DM)}}
 - **Diagnosis Threshold:** {{HbA1C level > 6.5% and <= 10.0%}}
 - **Medical Methodology:** {{Stable doses of oral anti-diabetic medications (<= 3) or lifestyle modification}}
 - **Optimization Target:** {{Pharmacokinetic evaluation, safety profiling, and reduction of hsCRP levels}}
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13. Study Procedures & Methodology

- **Management Pathway:** {{Daily oral administration of MK-2828 or placebo for 28 consecutive days}}
 - **Education Protocol (HCP):** {{Investigator training on Phase 1 safety monitoring and PK sampling windows}}
 - **Education Protocol (Patient):** {{Instructions for daily symptom logging and virtual visit adherence}}
 - **Quality Audit Mechanism:** {{Daily symptom tracking & study drug accountability monitoring}}
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14. Observation & Follow-Up Schedule

- **Total Duration:** {{~8 weeks (Screening: 2-4 wks, Treatment: 4 wks, Follow-up: 2 wks)}}
 - **Onsite Visit Cadence:** {{Screening, Day 1, Day 27, and post-treatment follow-up}}
 - **Remote Monitoring Frequency:** {{Daily virtual visits during the 28-day treatment phase}}
 - **Checklist Review Interval:** {{Baseline, Day 1, Day 27/28, and End of Study}}
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15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator		[Insert Name]		_____		[DD-MMM-YYYY]										
	Clinical Operations Lead		[Insert Name]		_____		[DD-MMM-YYYY]										
		Lead Statistician		[Insert Name]		_____		[DD-MMM-YYYY]									